MAT 201 - MATHEMATICAL METHODS I

LECTURE 1: REAL-VALUED FUNCTIONS OF A REAL VARIABLE

LECTURE C24-1-1 :: WED., NOV. 20, 2024 10:00 AM (WAT) - WED., NOV. 20, 2024 6:00 PM (WAT)

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OUTLINE

- Introduction
- Types of Functions
- Properties of Functions
- Limits of Functions
- Continuity
- Derivatives
- Conclusion

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INTRODUCTION: *DEFINITION*

- A real-valued function f of a real variable is a rule that assigns each real number x in a set D (the domain) to exactly one real number $f(x)$.

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INTRODUCTION: *FUNCTION NOTATION*

- We write $f : D \to \mathbb{R}$ to denote that f is a function from the domain D to the set of real numbers $\mathbb{R}$.
- If f is defined by an algebraic expression, we write $f(x) = \text{expression}$.

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INTRODUCTION: *DOMAIN AND RANGE*

- **Domain:** The domain of a function is the set of all possible input values (real numbers x for which $f(x)$ is defined).
- **Range:** The range of a function is the set of all possible output values (real numbers $f(x)$ for all x in the domain).

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INTRODUCTION: *GRAPHS OF FUNCTIONS*

- **Graphs:** The graph of a function is a visual representation of the relationship between the input and output values.
    - The x-axis represents the input values.
    - The y-axis represents the output values.

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INTRODUCTION: *TYPES OF FUNCTIONS*

- **Linear Functions:**
  $f(x) = mx + b$
  where m and b are constants.
- **Quadratic Functions:**
  $f(x) = ax^2 + bx + c$
  where a, b, and c are constants.

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PROPERTIES OF FUNCTIONS *EVEN AND ODD FUNCTIONS*

- A function f is even if $f(x) = f(-x)$ for all x in the domain.
- A function f is odd if $f(x) = -f(-x)$ for all x in the domain.

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LIMITS

The limit of a function $f(x)$ as x approaches a value c is denoted by
$\lim_{x \to c} f(x)$
and represents the behavior of the function near c.
If $\lim_{x \to c} f(x) = L$, it means that as x gets arbitrarily close to c, the values of $f(x)$ get arbitrarily close to L.

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CONTINUITY

A function f is continuous at a point c if
$\lim_{x \to c} f(x) = f(c)$
i.e., there are no abrupt jumps or holes in the graph of the function at $x = c$.

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DERIVATIVES

- The derivative of a function $f(x)$ with respect to x is denoted by $f'(x)$ and represents the rate at which the function is changing at a given point.
- Geometrically, the derivative corresponds to the slope of the tangent line to the graph of the function at a specific point.

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CONCLUSION

- Real-valued functions map real numbers to real numbers.
- Notation, domain, and range characterize a function.
- Graphs provide a visual representation.
- Different types and properties of functions.
- Limits, continuity, and derivatives offer insights into the behavior of functions.

LASU-VLAP

Your attendance for this lecture can not be marked at this moment because it was scheduled for Wed., Nov. 20, 2024 10:00 AM (WAT) - Wed., Nov. 20, 2024 6:00 PM (WAT) and you are outside the specified lecture period.

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